

Aegis-IoT Complete Execution Runbook

CE313 Final Project

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Purpose

This standalone document provides a complete step-by-step procedure to run the entire Aegis-IoT project, including model preparation, AnyLogic simulation, dashboard launch, federated learning, and database log verification.

1 Prerequisites

Ensure the following tools are installed:

- AnyLogic 8.x
- Python 3.8+
- Node.js 18+ and npm
- Java JDK 11+

Version checks:

```
python --version
node --version
npm --version
java -version
```

2 Install Dependencies

2.1 Python Dependencies

From project root:

```
pip install -r requirements.txt
```

2.2 Dashboard Dependencies

```
cd dashboard
npm install
cd ..
```

3 Database Logging Setup

The project database scripts include a centralized table named `RUNTIME_LOGS` to capture system logs in the database.

Updated files:

- `database/db.script`
- `Anylogic Models/FinalCCNProject1/database/db.script`

Verification query:

```
SELECT LOG_ID, LOG_TIME, LOG_LEVEL, SOURCE_COMPONENT, EVENT_TYPE, STATUS
FROM RUNTIME_LOGS
ORDER BY LOG_ID;
```

4 Train and Export ML Models

```
cd model
python train_latency_random_forest.py
python test_m2cgen4.py
cd ..
```

Expected outcome: predictor Java code is generated/updated for use in the AnyLogic runtime.

5 Inject Model Upgrades into AnyLogic Project

```
cd injection_scripts
python inject_multi_dashboard.py
python inject_rl_and_gan.py
cd ..
```

6 Run AnyLogic Simulation

1. Open AnyLogic.
2. Open `Anylogic Models/FinalCCNProject1/FinalCCNProject1.alpx`.
3. Run the main experiment.

Confirm the following:

- MQTT traffic generation starts.
- ML prediction and anomaly checks are active.
- RL policy updates are active.
- Dashboard telemetry stream is generated.

7 Run Web Dashboard

In a separate terminal:

```
cd dashboard
npm run dev
```

Open the Vite URL (typically <http://localhost:5173>).

8 Run Federated Learning

Start server:

```
cd federated
python cloud_server.py
```

Start edge nodes (one or more terminals):

```
cd federated
python edge_node.py
```

Generate convergence plot:

```
cd federated
python plot_federated.py
```

9 Final Validation Checklist

1. AnyLogic run completes without runtime failure.
2. Dashboard shows live metrics.
3. Federated scripts run and produce output plots.
4. MQTT_READY receives data.
5. RUNTIME_LOGS receives system log rows.

10 Quick Command Order

- 1) `pip install -r requirements.txt`
- 2) `cd dashboard && npm install`
- 3) `cd model && python train_latency_random_forest.py && python test_m2cgen4.py`
- 4) `cd injection_scripts && python inject_multi_dashboard.py && python inject_rl_and_g`
- 5) Run AnyLogic model: FinalCCNProject1.alpx
- 6) `cd dashboard && npm run dev`
- 7) `cd federated && python cloud_server.py`
- 8) `cd federated && python edge_node.py`
- 9) `cd federated && python plot_federated.py`